Enums

Example: Weekdays

```
public enum Day {
    MONDAY, TUESDAY, WEDNESDAY, THURSDAY, FRIDAY, SATURDAY, SUNDAY
}
```

```
public static void main(String[] args) {
    Day d = Day.FRIDAY;
    switch (r) {
        case MONDAY:
            System.out.println(“Monday! Bummer!”);
        // ...
        case FRIDAY:
            System.out.println(“Aww yeah!”);
        // ...
    }
}
```

Enumerations

We often have data that come in finitely many variants:

- Primary Colors: Red, Green, Blue.
- Operating Systems: Windows, Linux, Mac
- Compass Points: North, East, South, West.
- Deck of Cards: Ace, King, Queen, …
- Weekdays: Monday, Tuesday, Wednesday,…
- Planets: Mercury, Venus, Earth, Mars, Jupiter, Saturn, Uranus, and Neptune.
- …



(Pluto is not a planet)

With Interfaces and Classes

```
public interface Planet {
    String getName();
    double getMass();
}
```

```
public class Mercury extends Planet {
    public String getName() { ... }
    public double getMass() { ... }
}
```

```
public class Venus extends Planet {
    public String getName() { ... }
    public double getMass() { ... }
}
```

Example: Planets

```
enum Planet {
    MERCURY(1),
    VENUS(2),
    EARTH(3),
    MARS(4);

    private int position;

    Planet(int position) {
        this.position = position;
    }

    public int getPosition() {
        return position;
    }
}
```

Example: Planets

```
public static void main(String[] args) {
    Planet p = Planet.EARTH;
    switch (p) {
        case EARTH:
            int pos = p.getPosition();
            System.out.println(“It's earth! Our Homeworld.”);
            System.out.println(“The “ + pos + ” planet.”);
        default:
            System.out.println(“It is not earth.");
    }
}
```